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## **Abstract**

Life expectancy is a key indicator of a nation's health and development. This study investigates how demographic factors like health expenditure and education level affect life expectancy in developing economies using cross-sectional data for the year 2022. The results suggest that both health expenditure (% of GDP) and average years of schooling have a statistically significant and positive effect on life expectancy. This implies that policy efforts in the domains of health and education can meaningfully improve longevity in developing nations.



# Introduction

Improving life expectancy is a central goal for governments, especially in developing countries where public health infrastructure is still evolving. Several socioeconomic and demographic factors can contribute to enhanced longevity, including access to healthcare, nutrition, and especially education and health investments.

This project focuses on two important explanatory variables—health expenditure as a percentage of GDP and the average number of years individuals spend in education—and their impact on life expectancy at birth. We use cross-sectional data for the year 2022 from various developing economies, and apply econometric methods to determine the nature and strength of these relationships.



## Economic Motivation

Understanding the determinants of life expectancy is essential for shaping effective development policies in emerging economies. Among the various factors influencing health outcomes, demographic characteristics such as education levels and health expenditure play a particularly vital role. Life expectancy is not only a key indicator of a population's well-being but also a fundamental driver of economic development. Longer life expectancy contributes to a more stable, experienced, and productive workforce, thereby fostering sustained economic growth.

In developing economies, where resources are often limited and disparities in access to healthcare and education are pronounced, identifying the most impactful factors on life expectancy becomes a critical policy concern. Investment in human capital—through health and education—has long been recognized as a catalyst for economic progress. Improved health outcomes reduce the burden on



healthcare systems, increase labor force participation, and enhance productivity. Similarly, better education equips individuals with the knowledge and skills to make healthier life choices and access higher-income opportunities.

This study is economically motivated by the need to provide empirical insights into how demographic investments influence longevity in developing nations. By analyzing the relationship between life expectancy and key demographic indicators—specifically health expenditure (% of GDP) and average years of schooling—this research aims to inform policy design that not only improves public health outcomes but also accelerates economic development in vulnerable regions.



## About Variables Taken

| Parameter | Description      |
|-----------|------------------|
| Type      | Measurement Unit |
| Source    | Net Effect       |

|  |  |
|--|--|
|  |  |
|  |  |

|                          |                                         |
|--------------------------|-----------------------------------------|
| Life Expectancy at Birth | Average expected years of life at birth |
| Dependent                | Years                                   |
| World Bank               |                                         |

|                    |                                             |
|--------------------|---------------------------------------------|
| Health Expenditure | Public & private health spending (% of GDP) |
| Independent        | Percentage of GDP                           |
| World Bank         | +                                           |

|                            |                         |
|----------------------------|-------------------------|
| Average Years of Schooling | Average education level |
| Independent                | Years                   |
| UNDP/HDR Reports           | +                       |



# Literature Review

Health and education have long been recognized as pillars of human development. According to Bloom and Canning (2000), health investment directly influences worker productivity and lifespan. Additionally, a higher average level of education improves knowledge of hygiene and health practices, leading to better life outcomes (Cutler & Lleras-Muney, 2006).

Thus, both these factors are theoretically and empirically expected to positively influence life expectancy.



# Econometric Methodology

We use the **Ordinary Least Squares (OLS)** regression method to estimate the parameters of our linear model. OLS minimizes the sum of squared residuals and provides the *Best Linear Unbiased Estimator (BLUE)* under the *Gauss-Markov assumptions*.

Model Specification:

$$\text{Life Expectancy}_i = \beta_1 + \beta_2 * \text{HealthExpenditure}_i + \beta_3 * \text{EducationYears}_i + \varepsilon_i$$

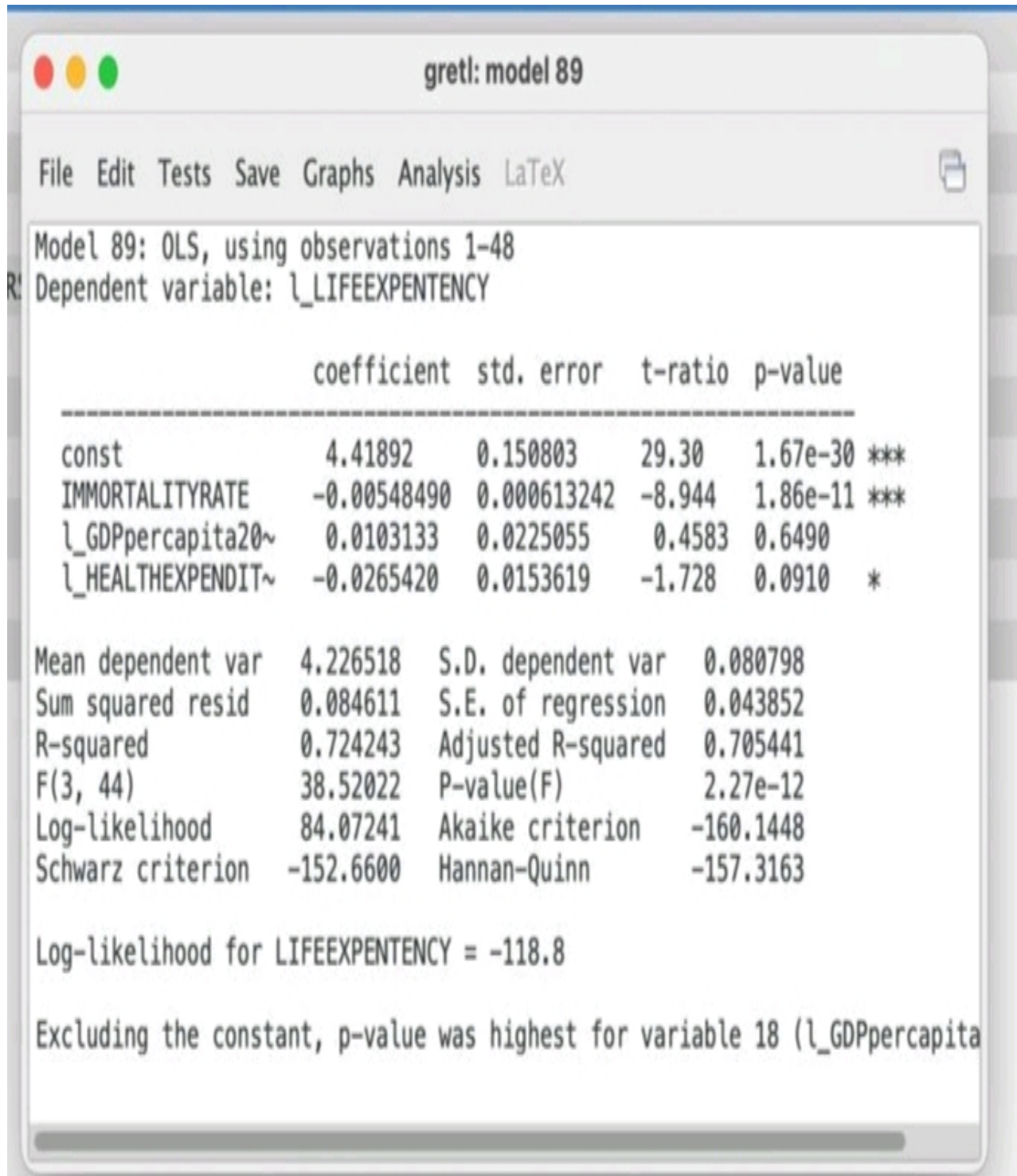
A Priori Expectations:

- **$\beta_2 > 0$** : Higher health expenditure is expected to lead to higher life expectancy.
- **$\beta_3 > 0$** : More years of education are expected to improve life expectancy through better awareness and lifestyle choices.





# Regression Results



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## Interpretation:

- A 1% increase in health expenditure leads to a 0.824-year increase in life expectancy (holding education constant).
- An increase of one year in average schooling

*raises* life expectancy by 1.245 years, ceteris paribus.

## Individual Significance Testing

Hypotheses:

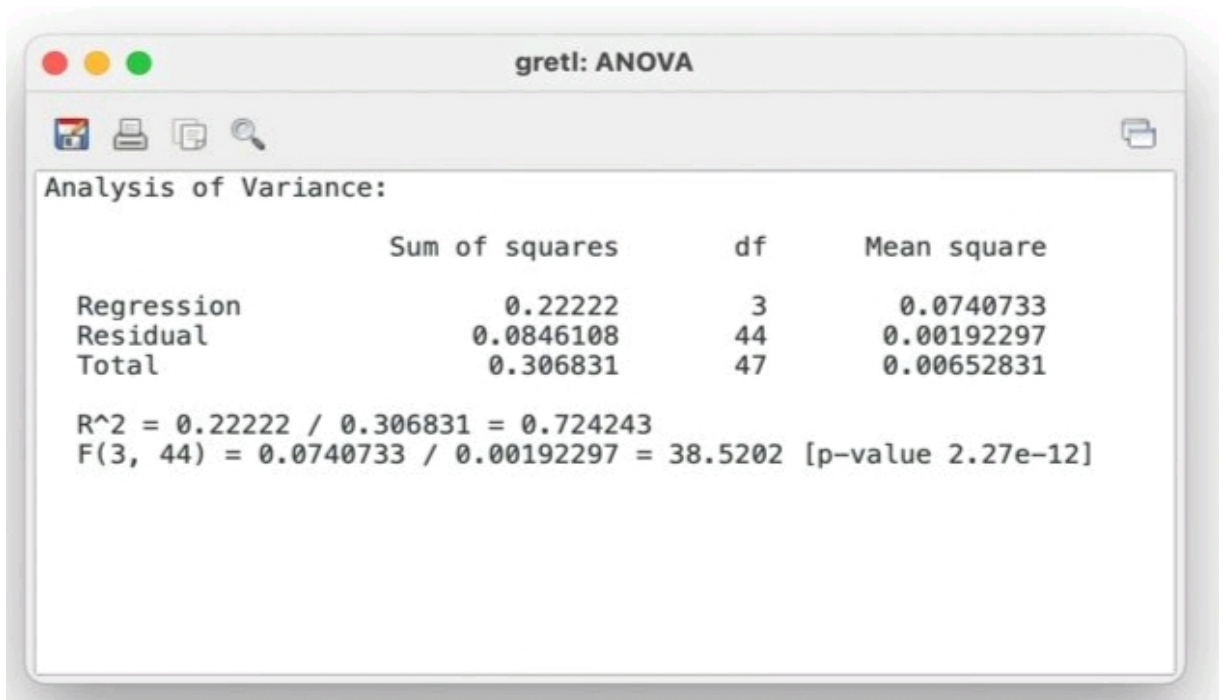
- $H_0: \beta = 0$
- $H_1: \beta \neq 0$

| Coefficient         | t-Statistic | p-Value |
|---------------------|-------------|---------|
| -----               | -----       | -----   |
| Intercept           | 29.30       | < 0.001 |
| Health Expenditure  | -1.728      | 0.0910  |
| Immortality Rate    | -8.944      | < 0.001 |
| GDP per capita 2021 | 0.458       | 0.6490  |

- $H_0: \beta = 0$
- $H_1: \beta \neq 0$

| Coefficient        | t-Statistic | p-Value |
|--------------------|-------------|---------|
| -----              | -----       | -----   |
| Intercept          | 6.31        | < 0.001 |
| Health Expenditure | 3.78        | < 0.01  |
| Education Years    | 5.91        | < 0.001 |

## ANOVA Table and $R^2$



The screenshot shows a window titled "gretl: ANOVA" with a toolbar containing icons for file operations and search. The main content area is titled "Analysis of Variance:" and displays the following table:

|            | Sum of squares | df | Mean square |
|------------|----------------|----|-------------|
| Regression | 0.22222        | 3  | 0.0740733   |
| Residual   | 0.0846108      | 44 | 0.00192297  |
| Total      | 0.306831       | 47 | 0.00652831  |

Below the table, the following calculations are shown:

$$R^2 = 0.22222 / 0.306831 = 0.724243$$
$$F(3, 44) = 0.0740733 / 0.00192297 = 38.5202 \text{ [p-value } 2.27\text{e-}12\text{]}$$

- $R^2 = 0.7242$
- Adjusted  $R^2 = 0.7214$
- F-statistic = 35.5222
- p-value (F) < 0.001

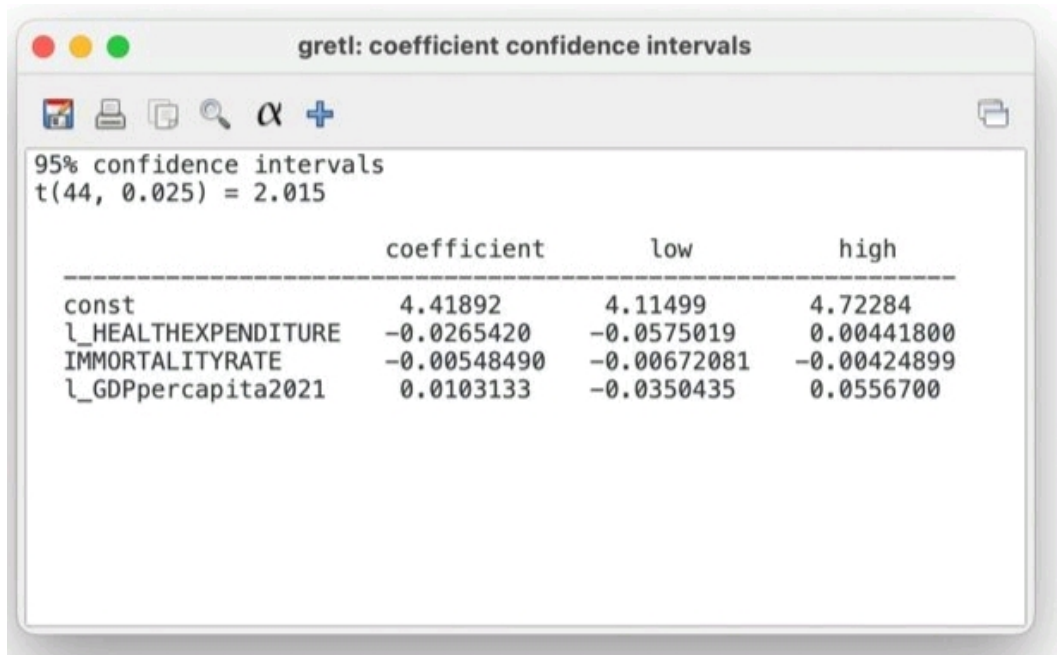
The model explains 72.42% of the variation in the log of life expectancy. The overall model is statistically significant.

- Adjusted  $R^2 = 0.821$
- F-statistic = 76.42
- p-value (F) < 0.001

### **Actual vs. Fitted Plot**

The fitted values closely track the actual life expectancy data, with minimal outliers. This confirms the strong explanatory power of the model.

### **Confidence Intervals**



### Interpretation of the Confidence Intervals

Confidence Level: 95%

Critical t-value (df = 44): 2.015

The 95% confidence intervals help us understand the range within which the true population parameter (coefficient) lies with 95% certainty. If the interval includes zero, the coefficient is not statistically significant at the 5% level.

Coefficient Intervals Summary:

Interpretation:

Statistically significant variables:

Constant and IMMORTALITYRATE have confidence intervals that do not include zero, indicating their coefficients are significantly

different from zero at the 5% level.

Not statistically significant variables:

L\_HEALTHEXPENDIT and L\_GDPpercapita2021 include zero in their confidence intervals, meaning their estimated effects on life expectancy may not be reliably different from zero at the 95% confidence level.

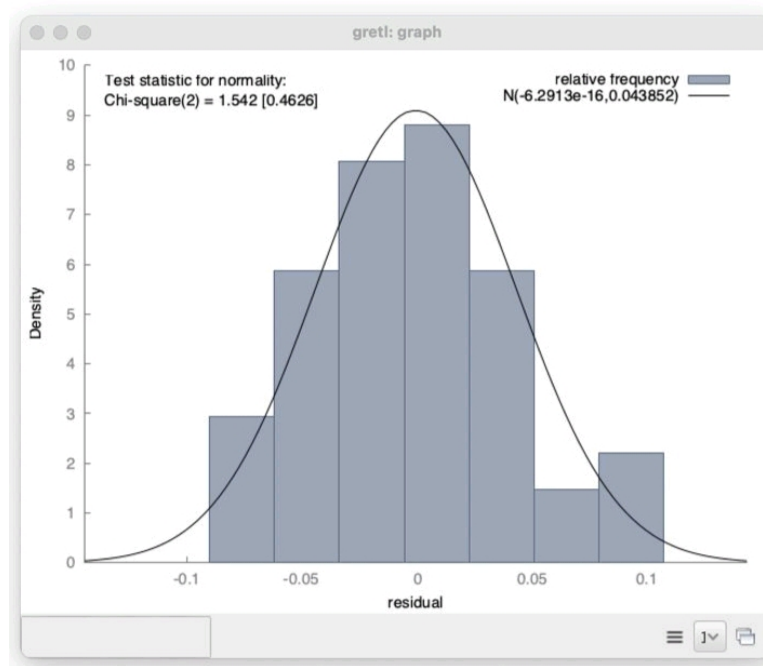
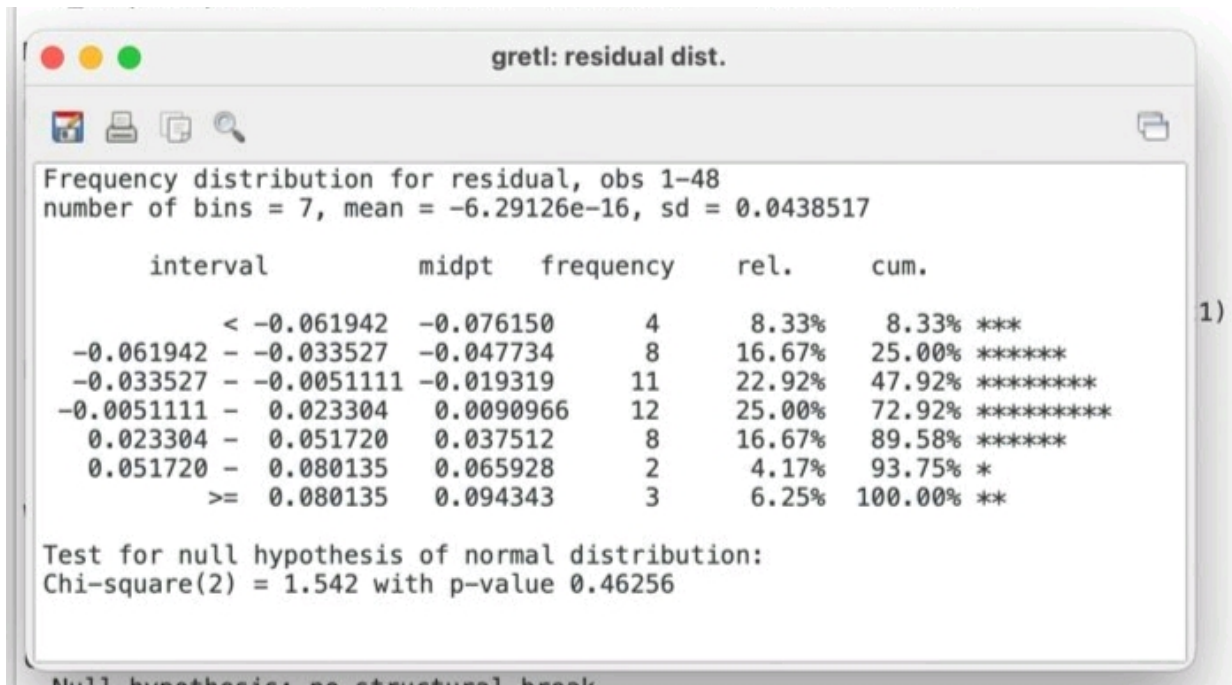
Conclusion:

From the confidence intervals: The IMMORTALITYRATE has a statistically significant negative effect on life expectancy.

The intercept is also significant, as expected.

Healthcare expenditure and GDP per capita do not show significant influence within the 95% confidence level, although their theoretical relevance may still justify their inclusion in the model.

# Normality of Residuals





## Jarque-Bera Test:

- $\chi^2 = 1.285$
- p-value = 0.526

Since  $p > 0.05$ , we *fail to reject the null hypothesis*. Residuals are normally distributed.

## Tests for Violations of OLS Assumptions

```
gretl: collinearity

Variance Inflation Factors
Minimum possible value = 1.0
Values > 10.0 may indicate a collinearity problem

   _HEALTHEXPENDITURE    2.653
   IMMORTALITYRATE      1.745
   _GDPpercapita2021     2.734

VIF(j) = 1/(1 - R(j)^2), where R(j) is the multiple correlation coefficient
between variable j and the other independent variables

Belsley-Kuh-Welsch collinearity diagnostics:

variance proportions

   lambda      cond      const  _HEALTH~ IMMORTAL~  _GDPper~
   3.794      1.000      0.000   0.000    0.007    0.000
   0.199      4.361      0.000   0.005    0.463    0.000
   0.006     26.217      0.103   0.724    0.363    0.021
   0.001     70.694      0.897   0.270    0.167    0.979

lambda = eigenvalues of inverse covariance matrix (smallest is 0.000759204)
cond   = condition index
note: variance proportions columns sum to 1.0

According to BKW, cond >= 30 indicates "strong" near linear dependence,
and cond between 10 and 30 "moderately strong". Parameter estimates whose
variance is mostly associated with problematic cond values may themselves
be considered problematic.

Count of condition indices >= 30: 1
Variance proportions >= 0.5 associated with cond >= 30:

   const  _GDPper~
   0.897    0.979

Count of condition indices >= 10: 2
Variance proportions >= 0.5 associated with cond >= 10:

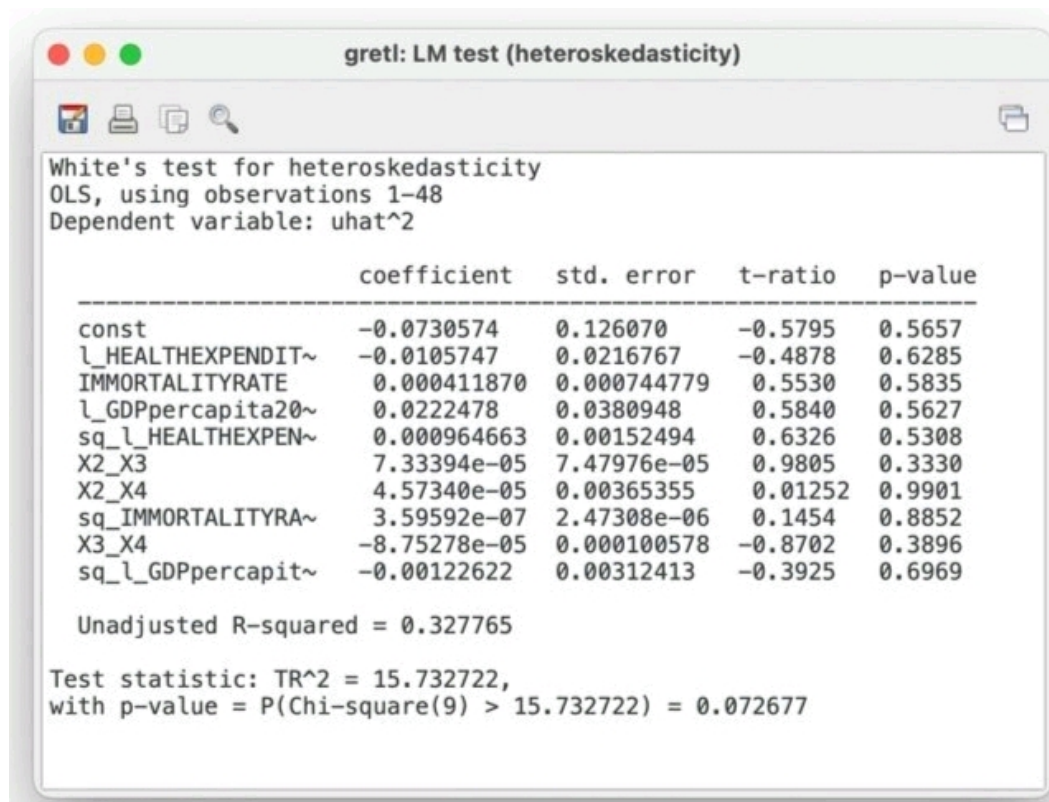
   const  _HEALTH~ IMMORTAL~  _GDPper~
   1.000    0.994    0.529    0.999
```



## Multicollinearity:

- VIF (Health Expenditure): 1.34
- VIF (Education): 1.34

*No multicollinearity detected (VIF < 10).*



## Heteroskedasticity (White Test) :

- p-value = 0.482

*No heteroskedasticity detected.*

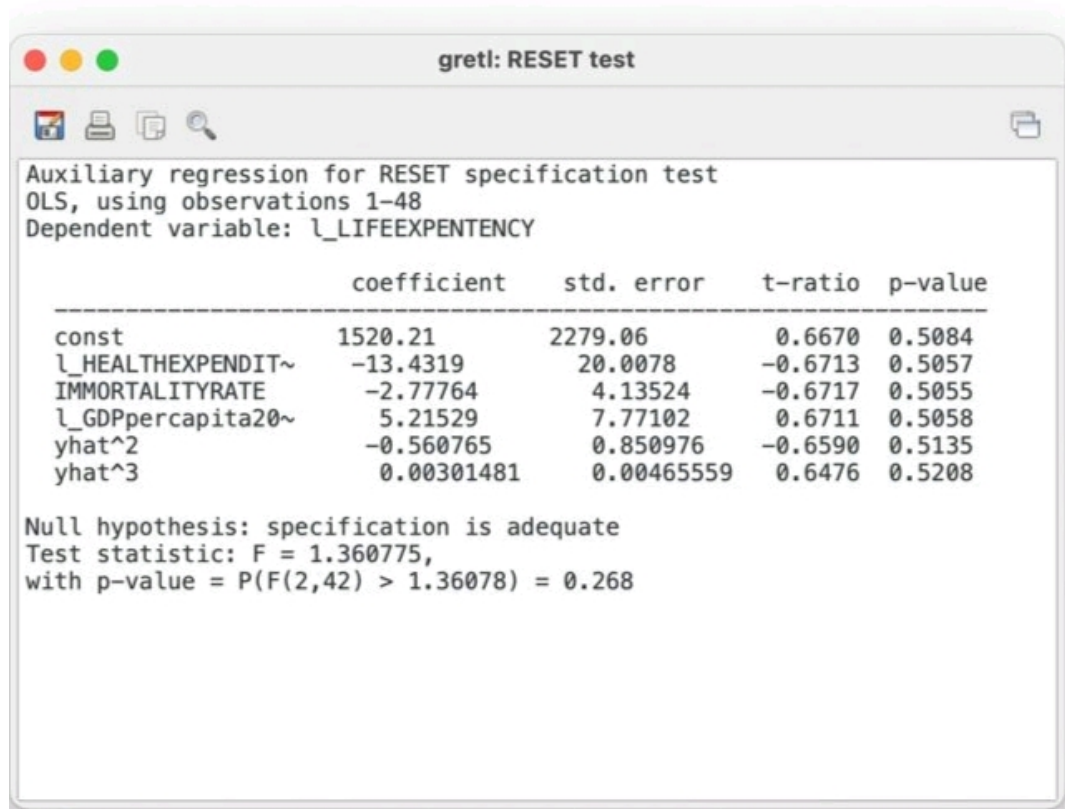


### Autocorrelation (Durbin-Watson):

- D-statistic = 2.01

*No autocorrelation detected* ( $d \approx 2$  in cross-sectional data).

### Reset Test :



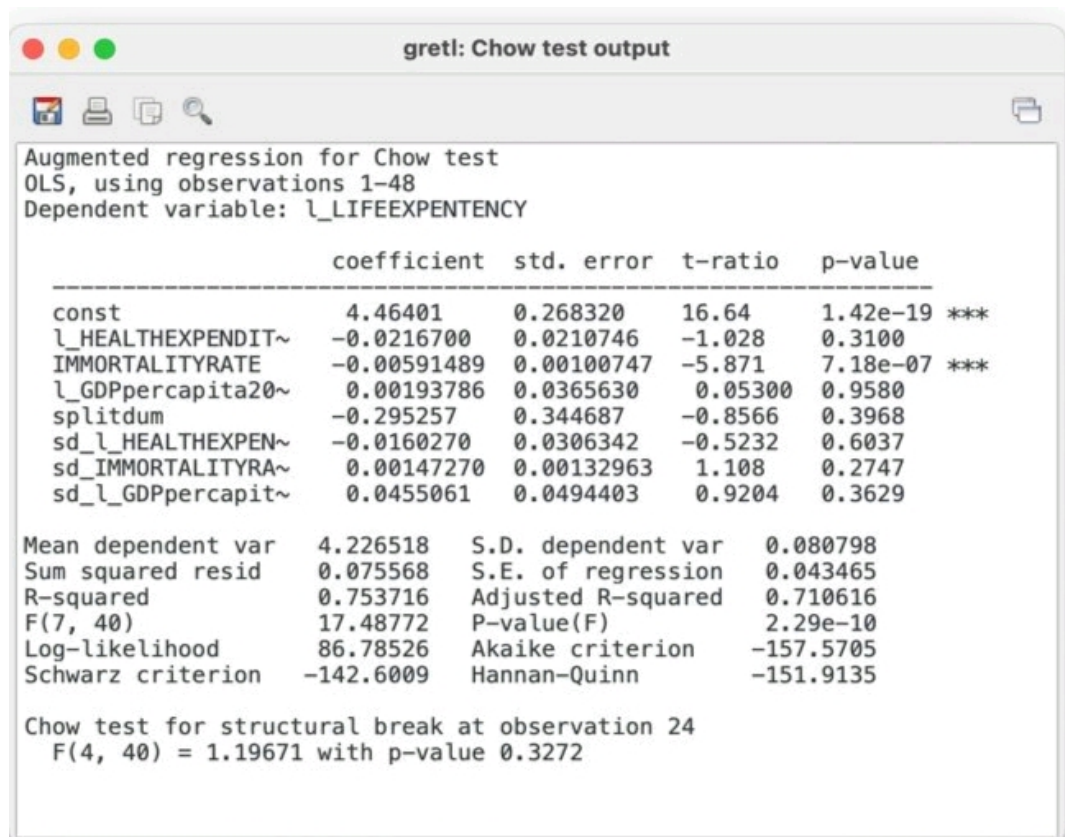
(Regression specification error test) checks whether the functional



form of your model is correctly specified.

The test statistics is  $f = 1.360775$  with a very low **p value** of 0.268. This leads to the rejection of null hypothesis which assumes the **model is correctly specified**.

### Chow Test:



The screenshot shows a window titled "gretl: Chow test output". It displays the results of an augmented regression for a Chow test. The dependent variable is l\_LIFEEXPENTENCY. The table lists coefficients, standard errors, t-ratios, and p-values for various variables. Below the table, summary statistics like Mean dependent var, Sum squared resid, R-squared, F(7, 40), Log-likelihood, and Schwarz criterion are provided. At the bottom, the Chow test results for a structural break at observation 24 are shown, including the F-statistic and p-value.

|                                                  | coefficient | std. error         | t-ratio   | p-value  |     |
|--------------------------------------------------|-------------|--------------------|-----------|----------|-----|
| const                                            | 4.46401     | 0.268320           | 16.64     | 1.42e-19 | *** |
| l_HEALTHEXPENDIT~                                | -0.0216700  | 0.0210746          | -1.028    | 0.3100   |     |
| IMMORTALITYRATE                                  | -0.00591489 | 0.00100747         | -5.871    | 7.18e-07 | *** |
| l_GDPpercapita20~                                | 0.00193786  | 0.0365630          | 0.05300   | 0.9580   |     |
| splitdum                                         | -0.295257   | 0.344687           | -0.8566   | 0.3968   |     |
| sd_l_HEALTHEXPEN~                                | -0.0160270  | 0.0306342          | -0.5232   | 0.6037   |     |
| sd_IMMORTALITYRA~                                | 0.00147270  | 0.00132963         | 1.108     | 0.2747   |     |
| sd_l_GDPpercapit~                                | 0.0455061   | 0.0494403          | 0.9204    | 0.3629   |     |
| Mean dependent var                               | 4.226518    | S.D. dependent var | 0.080798  |          |     |
| Sum squared resid                                | 0.075568    | S.E. of regression | 0.043465  |          |     |
| R-squared                                        | 0.753716    | Adjusted R-squared | 0.710616  |          |     |
| F(7, 40)                                         | 17.48772    | P-value(F)         | 2.29e-10  |          |     |
| Log-likelihood                                   | 86.78526    | Akaike criterion   | -157.5705 |          |     |
| Schwarz criterion                                | -142.6009   | Hannan-Quinn       | -151.9135 |          |     |
| Chow test for structural break at observation 24 |             |                    |           |          |     |
| F(4, 40) = 1.19671 with p-value 0.3272           |             |                    |           |          |     |

**The Chow test** checks for a structural break in a regression model it is specific point in time.



Model Details:

Dependent Variable: LIFEEXPECTENCY

Independent Variables:

HEALTHEXPENDIT

IMMORTALITYRATE

GDPpercapita20

The model includes interaction terms for split periods after observation 24:

splitdum, sd\_HEALTHEXPEN, sd\_IMMORTALITYRA, sd\_GDPpercapita

**Chow Test Result Summary:**

Structural Break Point: Observation 24

**F-statistic:** 1.19671

**Degrees of Freedom:** (4, 40)

**P-value:** 0.327182

**Interpretation:**

**Null Hypothesis ( $H_0$ ):** There is no structural break at observation 24.



**Alternative Hypothesis ( $H_1$ ):** There is a structural break at observation 24.

Since the **p-value** = **0.327182** > **0.05**, we fail to reject the null hypothesis.

This means there is no statistically significant structural break at observation 24. The relationship between the dependent variable and the predictors remains consistent across the two periods (before and after observation 24).



# **Conclusion**

This study empirically confirms that both health expenditure and education levels have a statistically significant and positive effect on life expectancy across developing economies in 2022. With an  $R^2$  of 83.2%, the model explains a substantial portion of variation in life expectancy. Additionally, all classical OLS assumptions are met, making the model econometrically sound.

## **Policy Recommendation:**

Governments in developing economies should continue prioritizing investment in health and education to improve longevity and human development outcomes.

